

Mihret Agegnehu Bekele

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SUMMARY

AI researcher with a strong interest in **Computer Vision**, particularly **multimodal learning**, **Vision-Language Models (VLMs)**, and **video understanding**. Passionate about developing AI systems for **low-resource languages and sign languages**, with a focus on building inclusive and accessible technologies. Also interested in **AI safety in vision systems**, particularly **identifying and mitigating racial and gender bias** and improving the reliability and fairness of computer vision models.

EDUCATION

Master of Science in Engineering Artificial Intelligence, Mastercard Foundation Scholar 08/2026 – 05/2028 | Kigali, Rwanda
Carnegie Mellon University Africa

Bachelors of Science in Software Engineering CGPA 3.75/4 06/2020 – 06/2025 | Addis Ababa, Ethiopia
Addis Ababa Science and Technology University 

PUBLICATIONS & MANUSCRIPTS

- [1] Damaris Stephanie Ndjebayi, **Mihret Agegnehu Bekele**, Anteneh Yehalem Tegegne, Yohannes Ayana Ejigu, Tamiru Alemnew. "Continuous Ethiopian Sign Language Recognition: Video and Skeleton-Based Models with Cross-Lingual Transfer" NeurIPS WiML, 2026.
- [2] **Mihret Agegnehu Bekele**. "Multimodal Video Understanding for Predicting and Optimizing Short-Form Content Engagement"  "CVPR Demo track, 2026.
- [3] **Mihret Agegnehu Bekele**, Dawit Getahun Mangistu, Kidus Paulos Gebresadik, Natnael Abayneh Unasho, Amanuel Gizachew Abebe, Simret A Gebreegziabher. "Steering Vision Models towards Subjective Concepts with Language"  "ACM CHI BiAlign Workshop, 2026.

RESEARCH EXPERIENCE

Carnegie Mellon University Africa  09/2026 – Present | Kigali, Rwanda
Research Assistant

- **Reproduced and benchmarked state-of-the-art OCR approaches** on the Fidel dataset for historical African manuscripts, evaluating existing models and analyzing their performance on low-resource writing systems.
- **Developed and evaluated OCR models for historical document recognition**, investigating data preparation, model training, and evaluation strategies for recognizing under-resourced African/Ethiopian scripts.
- **Designed an open OCR benchmarking competition** based on the Fidel dataset, developing starter notebooks and standardized evaluation workflows to enable researchers to develop and compare models beyond established baselines.

Africa Computer Vision Summer School (ACVSS)  07/2026 – 07/2026 | Accra, Ghana
Student, Google

- **Relevant Coursework:** Introduction to Computer Vision, Advanced Architectures for Computer Vision, Fairness and Ethics, Generative Modeling, Multimodal Vision, World Models, Vision Robotics, 3D Vision, Generative Models and Diffusion Models.
- **Speakers and Organizers:** David Forsyth, Awa Dieng, Adam Kortylewski, Ronald Clark, Amir Zamir, Deepti Ghadiyaram, Amr Khalifa, Deborah D. Kanubala, Max Welling, Abigail Annkah, Aisha Walcott-Bryant, Michelle Lochner, Sekou Remy, Shreyank Gowda, Jean-Michel A Sarr, Abigail Oppong, Stella Yu, Raesteji, Sefala, Raoul de Charette, Anurag Arnab.

Computer Vision and Graphics Research Initiative  12/2025 – 01/2026 | Addis Ababa, Ethiopia
Research Project: Steering Vision Models Subjective Concepts

- Selected for a competitive computer vision research program focused on computer vision fundamentals.
- Conducted research  on **vision-language models (CLIP)** to enable inference-time learning of subjective, user-specific visual concepts using natural-language critiques.
- Designed and Developed  a co-adaptive query update mechanism to improve visual grounding without model fine-tuning.
- **Mentored by:** Simret Gebreegziabher, PhD Student, University of Notre Dame.

Addis Ababa Science and Technology University  01/2024 – 06/2024 | Addis Ababa, Ethiopia
Research Project: AI Assisted Resource Administering and Allocation

- Designed and implemented an AI-based exam scheduling system using **Particle Swarm Optimization (PSO)**, incorporating accessibility-aware constraints and unstructured data processing.
- Improved exam room utilization efficiency by **40%** and presented results at an online research conference.
- **Supervised by:** Dr. Surafel Tilahun, Associate Professor, Addis Ababa Science and Technology University.

PROFESSIONAL EXPERIENCE

ChromaWay 

Junior Software Developer

08/2025 – 08/2026 | Stockholm, Sweden

- **Deployed** the Colorpool DEX mainnet backend on **AWS (EC2, RDS PostgreSQL, ElastiCache Redis with TLS/stunnel)**, running a Dockerized API and 6+ PM2 worker processes behind nginx.
- Developed **clawchain.ai**  a blockchain-powered, Reddit-like platform for AI agents using **Open Claw**.
- **Improved analytics accuracy by 18%** by identifying tracking gaps through Hotjar integration and Matomo opt-out analysis, uncovering inconsistencies in user behavior data on Chromia Docs.

Software Developer Intern

02/2025 – 08/2025 | Stockholm, Sweden

- Developed a full-stack project in **1 week** while learning a new language, Rell, earning top **2% ranking**.
- Integrated a **recommendation algorithm** into a social media platform to deliver more relevant content.

- Deployed applications on **GCP**, leveraging **Google Cloud Storage** and **Firestore** to efficiently store and manage application metrics.
- Reduced task classification time by **60%** through **multithreading** optimization in AI orchestration pipeline.
- Improved function-calling accuracy by **40%** by optimizing schemas and prompts.
- Implemented unit and integration tests, increasing test coverage by **35%** and ensuring system stability.

TEACHING EXPERIENCE

- Delivered daily mini-lectures to reinforce complex lab topics and boost student engagement on Data Structures & Algorithms.
- Provided one-on-one support during and after class to help struggling students catch up.
- **Professors:** Jelani Nelson (UC Berkeley/Google), Daniel Kang (UIUC), Huy Nguyen (Northeastern), Huacheng Yu (Princeton)

PROJECTS

Sign to Learn

08/2026 – Present

- **Developed** an offline-first educational platform leveraging computer vision to evaluate sign-language video responses in real time for deaf and hard-of-hearing learners.
- **Developed a lightweight computer vision pipeline** utilizing MediaPipe keypoint tracking and Dynamic Time Warping (DTW) on a initial math digit-recognition scope to optimize real-time gesture evaluation.
- **Architected a cross-platform desktop application** in Flutter optimized for low-bandwidth and offline environments, enabling local data storage and zero-connectivity assessment workflows.

Restoring Ethiopic Manuscripts for Document Understanding

05/2026 – Present

- Supervised by **Prof. Maarten de Rijke**.
- Developed an AI-based pipeline for the digital restoration and understanding of damaged Ethiopic (Ge'ez) manuscripts using computer vision, OCR, and multimodal generative models.
- Fine-tuned **YOLO** on a synthetically generated dataset of damaged manuscript images to detect and localize regions requiring restoration.
- Integrated **OCR-assisted damage localization** with vision-language models to infer missing or damaged textual content from the visual and linguistic context of manuscript pages.
- Explored diffusion-based and autoregressive approaches for patch-level manuscript restoration.
- Built data-processing and experimentation pipelines using **Python, OpenCV, PIL, and PyTorch, Runpod GPU**.

Multimodal Video Understanding for Predicting and Optimizing

05/2025 – Present

Short-Form Content Engagement | Research Project

- **Accepted to CVPR 2026 Demo**.
- **Built a multimodal video pipeline** combining text, audio, visual, and **deep learning features (CLIP, Wav2Vec2, Sentence Transformers)** to rank high-engagement segments.
- **Applied classical + deep CV techniques** (Canny, Sobel, histogram scene detection, CLIP) with **7-mode ablation studies** to measure modality impact.
- **Engineered a scalable pipeline** using **multi-processing, multi-threading, caching, and structured logging** for efficient analysis.

VOLUNTEER EXPERIENCE

Develop for Good | Product Lead

03/2025 – 02/2026 | California, USA

- Managed the end-to-end selection process by reviewing **350+ applications** to staff high-performing teams for **4** nonprofit partnerships.
- Directed **40+ personnel** across 4 projects—including **3 managed concurrently**—leading project managers, technical managers, mentors, designers, and engineers to ensure high-quality delivery for all clients.

Develop for Good | Technical Manager

10/2024 – 02/2025 | California, USA

- Reduced manual mentor-matching time by **100%** by implementing an algorithm for CovEducation's platform, improving mentorship quality for over **7,000 K-12** students.
- Led a team of **6** engineers to develop secure, stable software by interpreting architecture, design, and nonfunctional requirements.

ACHIEVEMENTS AND CERTIFICATES

- **2nd Place, Africa Computer Vision Summer School Hackathon:** Continuous Ethiopian sign language recognition, **hosted at Google**.
- **Very Great Distinction:** Achieved **top 5%** at Addis Ababa Science and Technology University Software Engineering Department.
- **A2Svian of the Year:** Achieved **1st place out of 60** people for my skills in **Data Structures and Algorithms**.
- **Technical Lead, Google Developer Group:** Directed a flutter curriculum for **100+ students**, coordinating **7 mentors**.

TECHNICAL SKILLS

Frameworks and Libraries: .NET, Firebase, Typescript, React, Go Gin, Docker, Flutter, Flask.

Programming Languages: C++, Go, Python, Java, C#, JavaScript, Dart

Machine Learning & Computer Vision: PyTorch, OpenCV, scikit-learn, NumPy, Pandas, Hugging Face Transformers, Vision-Language Models (CLIP), Faster-Whisper, YOLO, Scrapy, Selenium, DBT, BeautifulSoup.

Databases: MongoDB, MySQL, Postgres.